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# Transformers Networks applied to Natural Language Processing (NLP)

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What is Natural Language Processing  
(NLP)?



# What is Natural Language Processing (NLP)?

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- Most of the first solutions on NLP were based on **hand written** rules
- As Machine Learning (ML) models gained popularity, a few models were used on NLP
  - Machine Learning (ML), is defined as the set of techniques used by a computer to **learn** from data
  - Some popular ML models used on NLP
    - Naive Bayes
    - Support Vector Machines
    - Maximum Entropy
    - **Neural Networks/Deep Neural Networks**

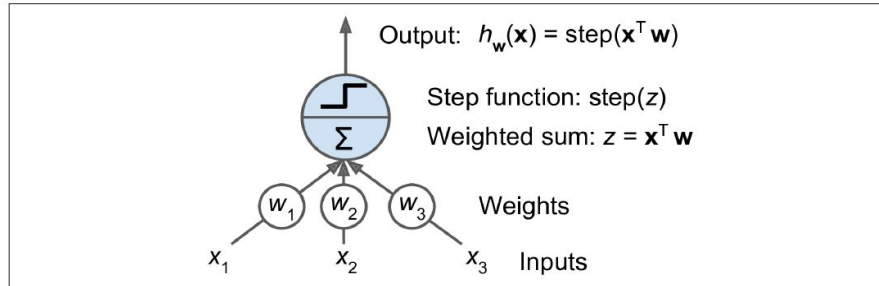


Source: "Words." by ninachildish is marked with CC BY-ND 2.0.

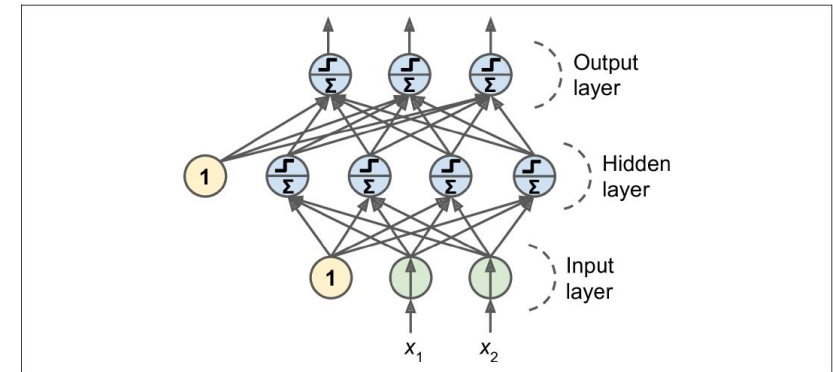
# Neural Networks/Deep Neural Networks

# Neural Networks/Deep Neural Networks

- **Neural Networks (NN)** are a set of Machine Learning models **inspired** on the human brain.
- The building blocks of NN are **the neurons**. Neurons can be connected.
- NN can be split in sub groups of neurons called **layers**.
- A NN with a large number of layers is called a **Deep Neural Network (DNN)**



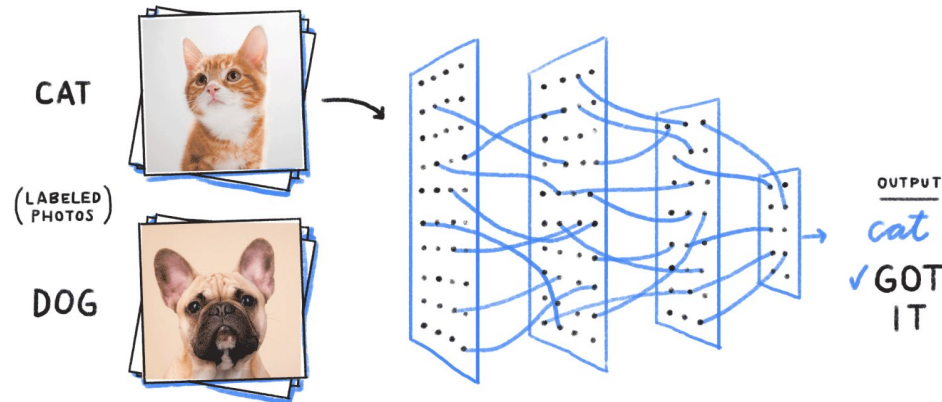
A single neuron. "Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow"



A Neural Network. "Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow"

# Neural Networks/Deep Neural Networks

- Neural Networks can **learn** certain tasks by **training**.
  - On each **training step**, you show an input and the expected result for that input.
  - For each error, you have adjust the weights until you have the desired result.

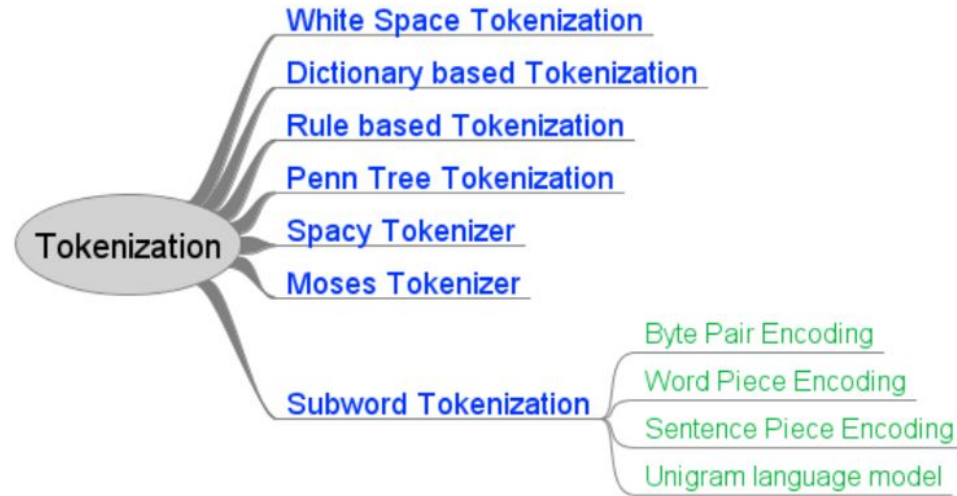


# Deep Learning on NLP



# Deep Learning on NLP - Tokenization

- First, we need to decide how we will divide our text to process it. This is called **tokenization**.
- There are different kinds of tokenization

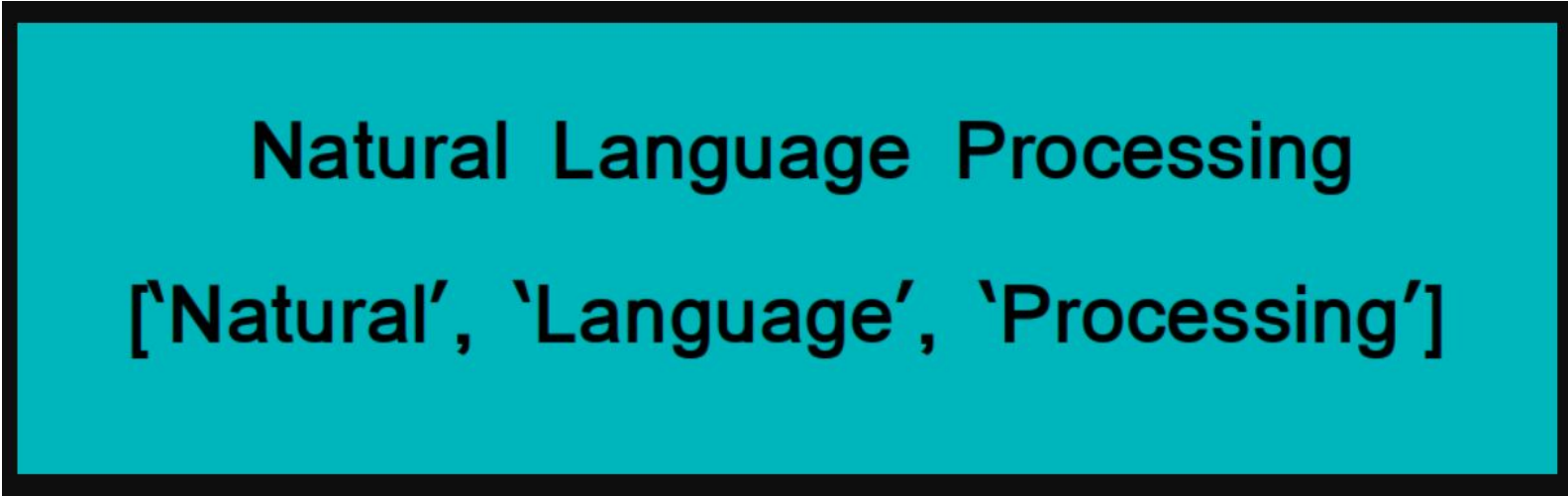


Source: <https://towardsdatascience.com/tokenization-for-natural-language-processing-a179a891bad4>

# Deep Learning on NLP - Tokenization

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- Example: White Space Tokenization



**Natural Language Processing**  
**['Natural', 'Language', 'Processing']**

Source:  
<https://www.analyticsvidhya.com/blog/2019/07/how-get-started-nlp-6-unique-ways-perform-tokenization/>

# Deep Learning on NLP - Embedding

- Deep Neural Networks are not compatible with “words”.
- We need to translate words to numbers. Some techniques are:
  - Replacement: Replace a word with a number

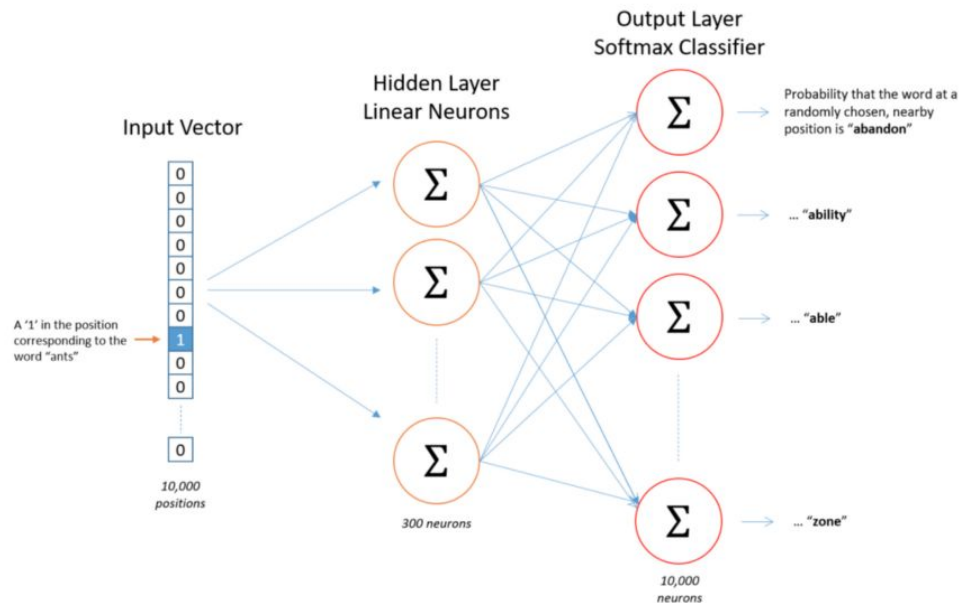
| I | like | writing | stories | on | the | computer. |
|---|------|---------|---------|----|-----|-----------|
| 1 | 2    | 3       | 4       | 5  | 6   | 7         |

- One Hot Encoding: Replace a word with a vector. The vector contains information about word presence.

|           | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|-----------|---|---|---|---|---|---|---|
| I         | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| like      | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| writing   | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| stories   | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| on        | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| the       | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| computer. | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

# Deep Learning on NLP - Embedding

- Deep Neural Networks are not compatible with “words”.
- We need to *translate words to numbers*. Some techniques are:
  - Embedding: Replace a word with a vector with **information** about the word and its **relation** with other words. (E.g. *Word2Vec*)



Check <https://projector.tensorflow.org/>

# Deep Learning on NLP - Processing

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- Now, with this new representation, we can start **understanding** each phrase.
- The information from a phrase does not only rely on the individual meaning of each word, but also on its **context**.

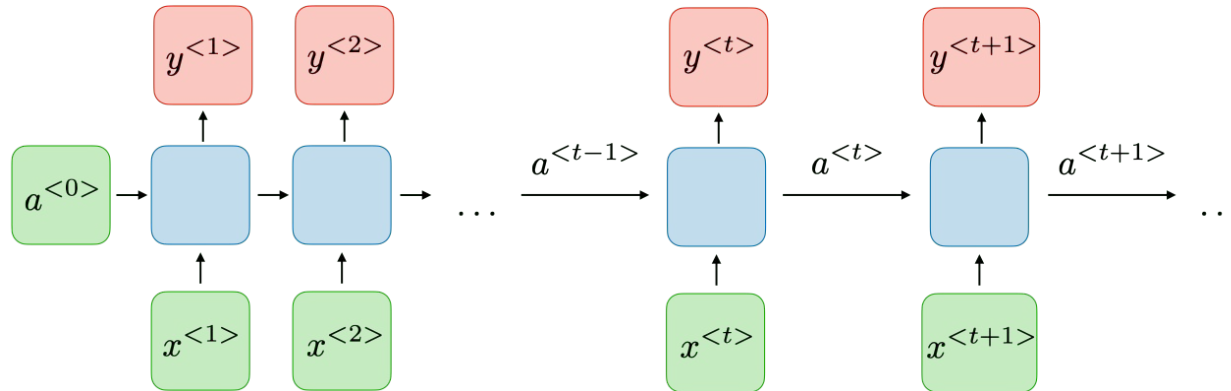
The apple is **red**.

The apple is **not red**.

- One approach is using **Recurrent Neural Networks (RNN)**

# Deep Learning on NLP - RNN

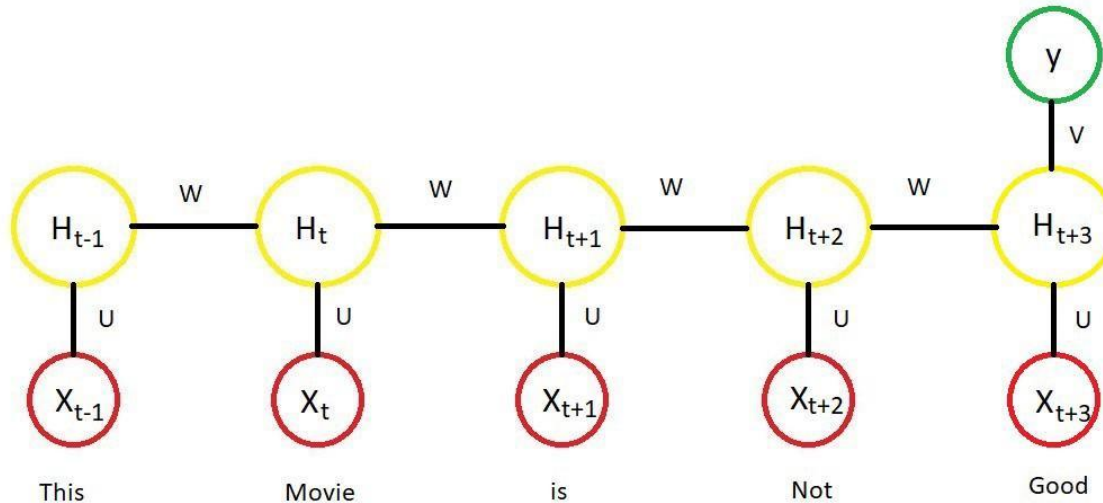
- A **Recurrent Neural Network (RNN)** are used mostly for sequential data.
- Each output neuron depends not only on the present input, but also on what happened on the last training step.



Source: <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks>

# Deep Learning on NLP - RNN

- Types of RNN architectures
  - Many to One: Multiple inputs are used to obtain one output. E.g. classification

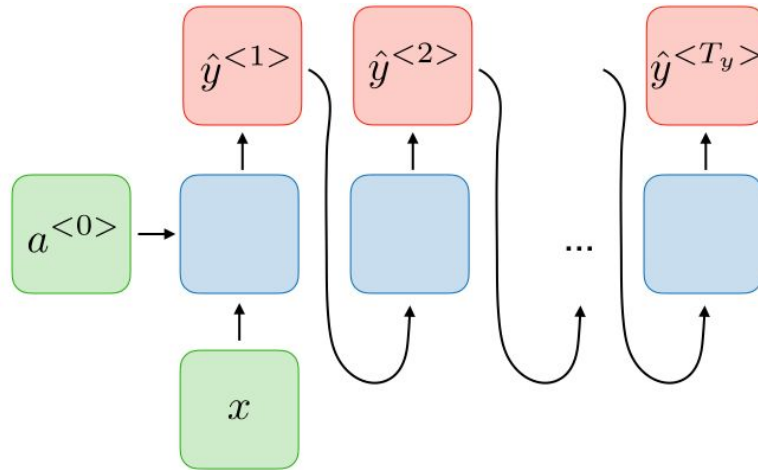


Source:

<https://medium.com/analytics-vidhya/natural-language-processing-from-basics-to-using-rnn-and-lstm-ef6779e4ae66>

# Deep Learning on NLP - RNN

- Types of RNN architectures
  - One to Many: A single input produces multiple outputs. E. g. Music generation



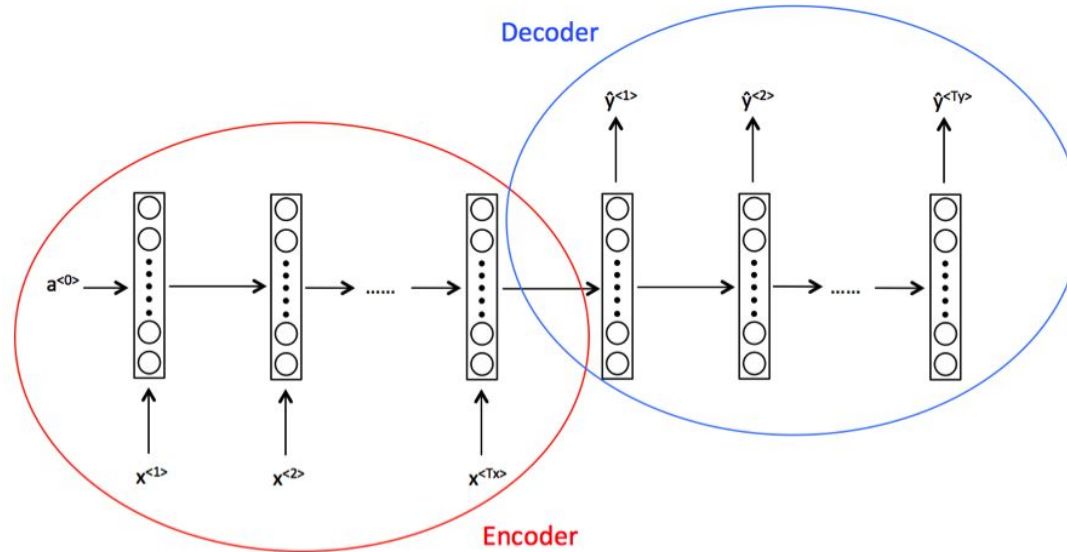
Source:

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# Deep Learning on NLP - RNN

- Types of RNN architectures
  - Many to Many: Multiple inputs produce multiple outputs. E. g. Translation



Source:

<https://medium.com/analytics-vidhya/natural-language-processing-from-basics-to-using-rnn-and-lstm-ef6779e4ae66>

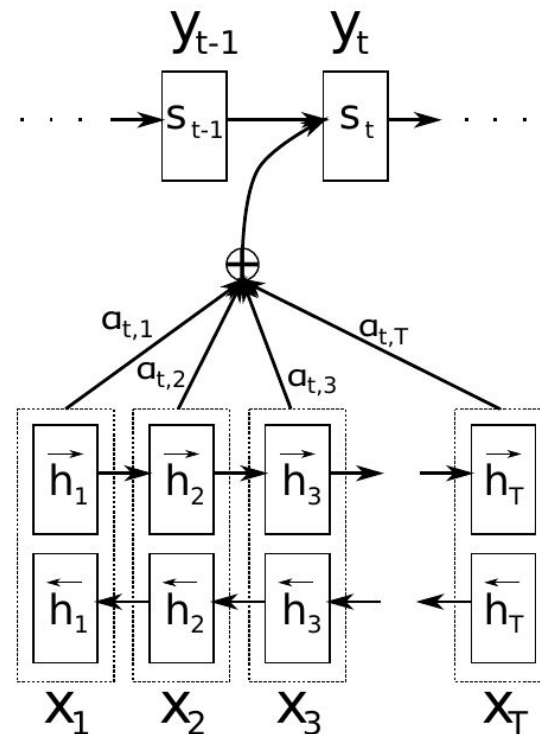
# Deep Learning on NLP - RNN

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- Limitations of RNN:
  - Dependency on data can be possible in only one direction: only data before can affect data later.
  - RNN are not capable to capture long term dependencies.
- In order to improve, we have to put more attention!

# Deep Learning on NLP - RNN and Attention

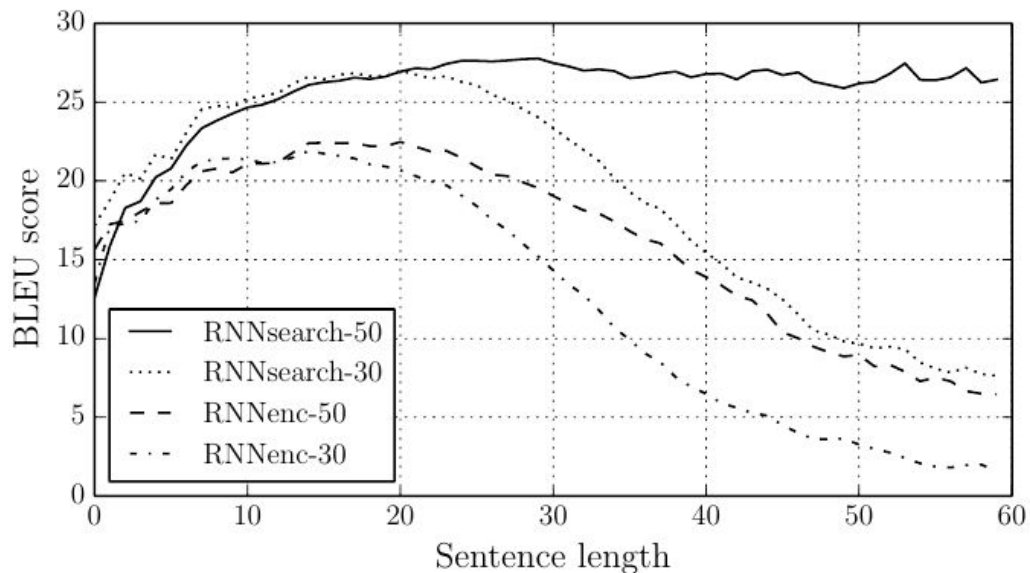
- Attention mechanisms are able to include information of the whole context to each iteration of RNN training process.
- For including context, we have to learn the importance of each input on the whole context.
- This importance is learned using a bidirectional RNN where the outputs are weights.



Source: Bahdanau, D., Cho, K., and Bengio, Y., "Neural Machine Translation by Jointly Learning to Align and Translate", arXiv e-prints, 2014.

# Deep Learning on NLP - RNN and Attention

- Attention mechanism provided incredible results on NLP.

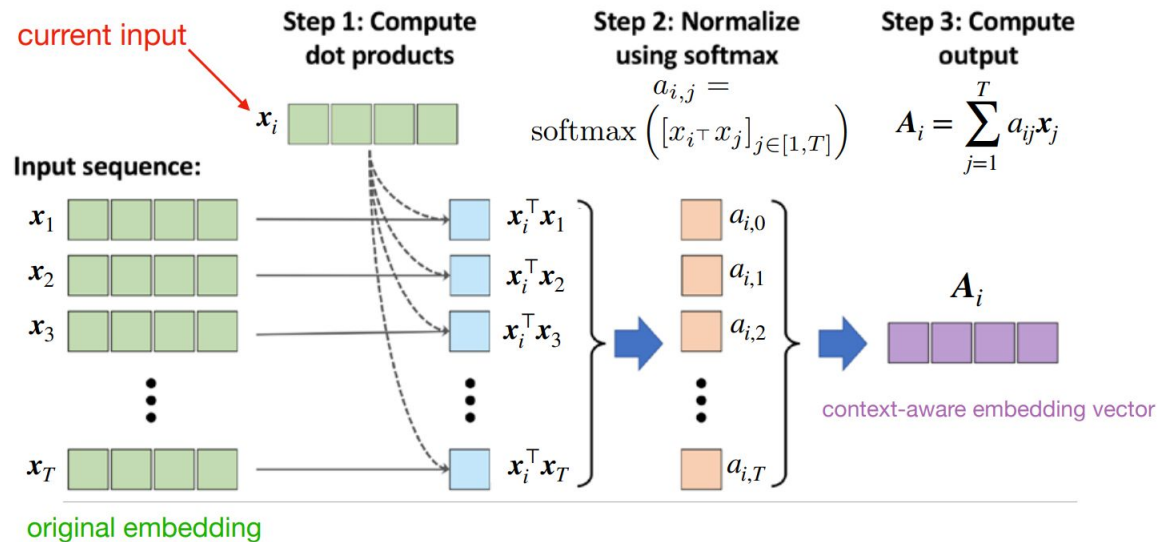


Source: Bahdanau, D., Cho, K., and Bengio, Y., "Neural Machine Translation by Jointly Learning to Align and Translate", arXiv e-prints, 2014.

- As you can see, Attention mechanism provides an incredible improvement on RNN scores, but... do we really need RNN? **Attentions is all you need**

# Deep Learning on NLP - Attention is all you need

- **Self-attention** is an attention mechanism for finding relations between different input position data from a single sequence.
- Problem: no trainable parameters



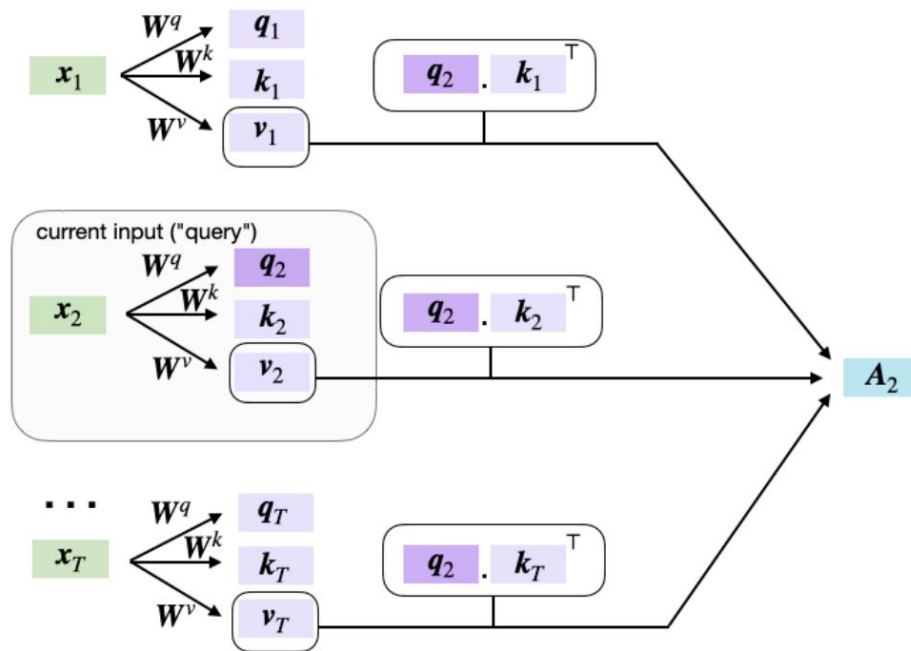
# Deep Learning on NLP - Attention is all you need

- **Scale dot-product attention** uses three different matrices called **query**, **key** and **value**. Those matrices can be trained

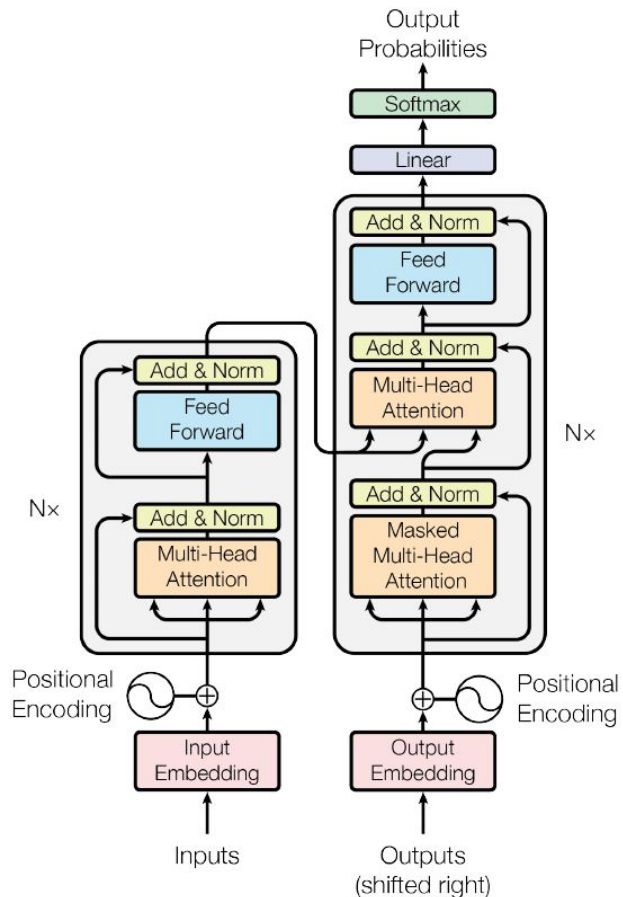
$$\text{query} = W^q x_i$$

$$\text{key} = W^k x_i$$

$$\text{value} = W^v x_i$$



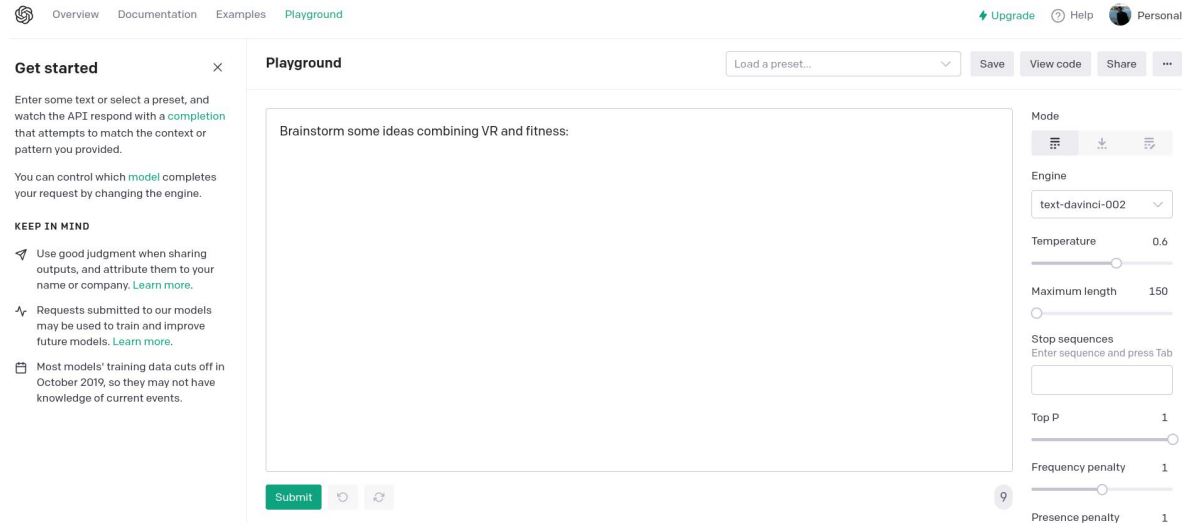
# Deep Learning on NLP - Attention is all you need



- **Transformers** are an architecture that use multi-head attention systems
- Includes multiple layers of encoders and decoder.
- A positional encoder is included in order to maintain information about position.

# Deep Learning on NLP - Applications

- Github copilot: <https://copilot.github.com/>
- Compose.ai (writing extension): <https://www.compose.ai/>
- Playground: <https://beta.openai.com/playground/>
- Etc...





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**Thank you for your attention!**

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# References

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